Ndumiso Baliti **28 March 2026**

BLTNDU001 **Prof. M. Vichi**

SEA4001W

Data Science

P4: Final Assignment

1. **Introduction**

I chose the study area between **17°E and 20°E, and 35°S and 32°S** on the west coast of South Africa because it lies within the southern Benguela upwelling system, which is known for being one of the most productive coastal regions. This area experiences strong seasonal upwelling, so it is a good region to study chlorophyll-a variability. The bathymetry in the region includes the coastal shelf and slope, which are important physical features as they help explain the relationship between the dept and the chlorophyll-a distribution in the ocean.

2. **Datasets**

The chlorophyll data is a global monthly climatology from the ESA-CCI Ocean Colour product. The dataset has a high spatial resolution of about 4 km. The variable chlor_a represents chlorophyll-a concentration at the ocean surface and is used as a proxy for phytoplankton biomass.

ESA-CCI. Ocean Colour Climate Change Initiative Dataset.

The bathymetry dataset is a regional GEBCO_2025 grid for the study area between 17°E–20°E and 35°S–32°S. It contains a single elevation variable on a 720 × 720 latitude-longitude grid, with a medium resolution. The values represent seabed elevation and topography, so negative values indicate ocean depth below sea level.

GEBCO (2025). GEBCO_2025 Grid.

3. **Plots**

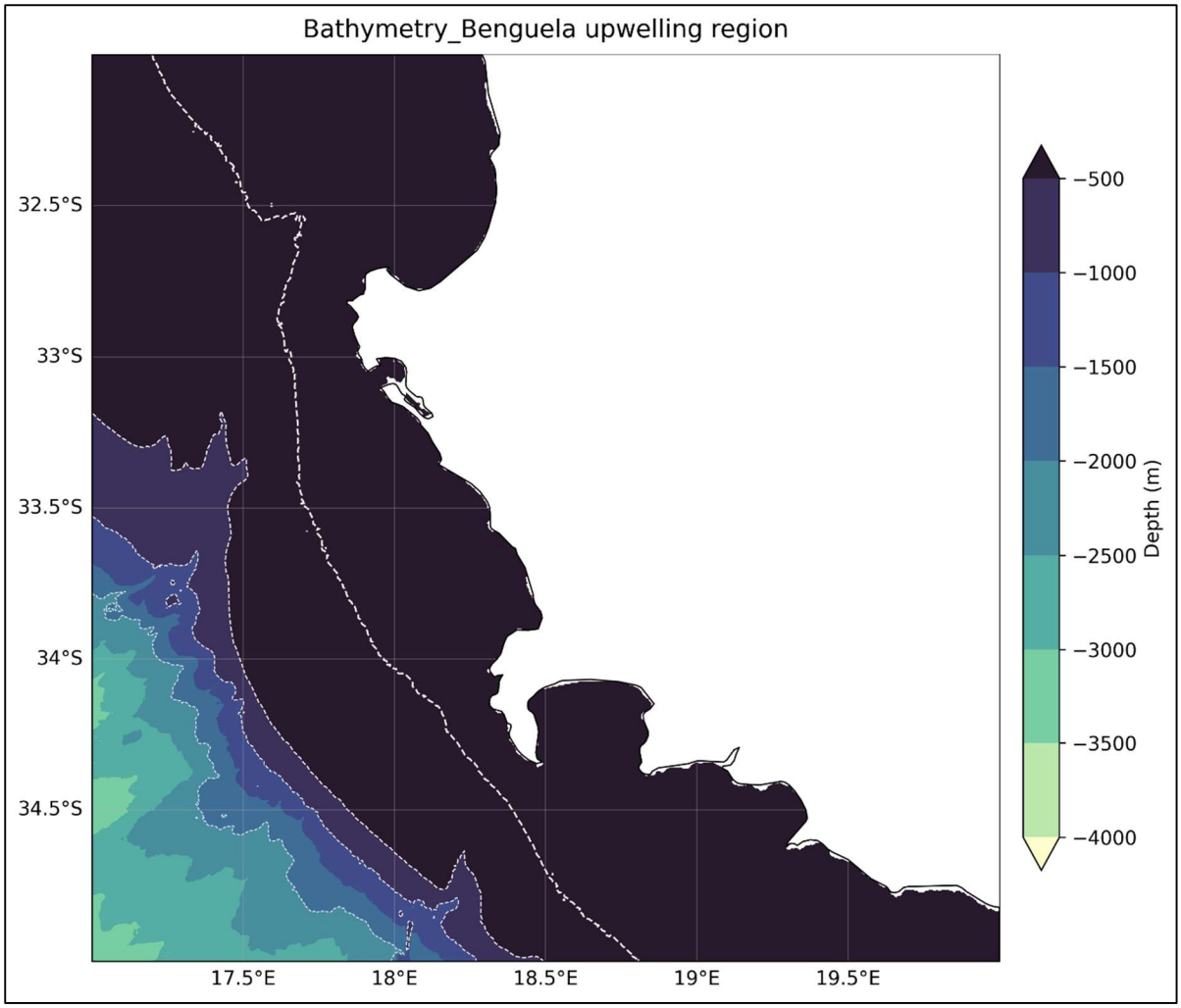

Figure 1: Bathymetry of the study area within the Benguela upwelling system on the southwest coast of South Africa.

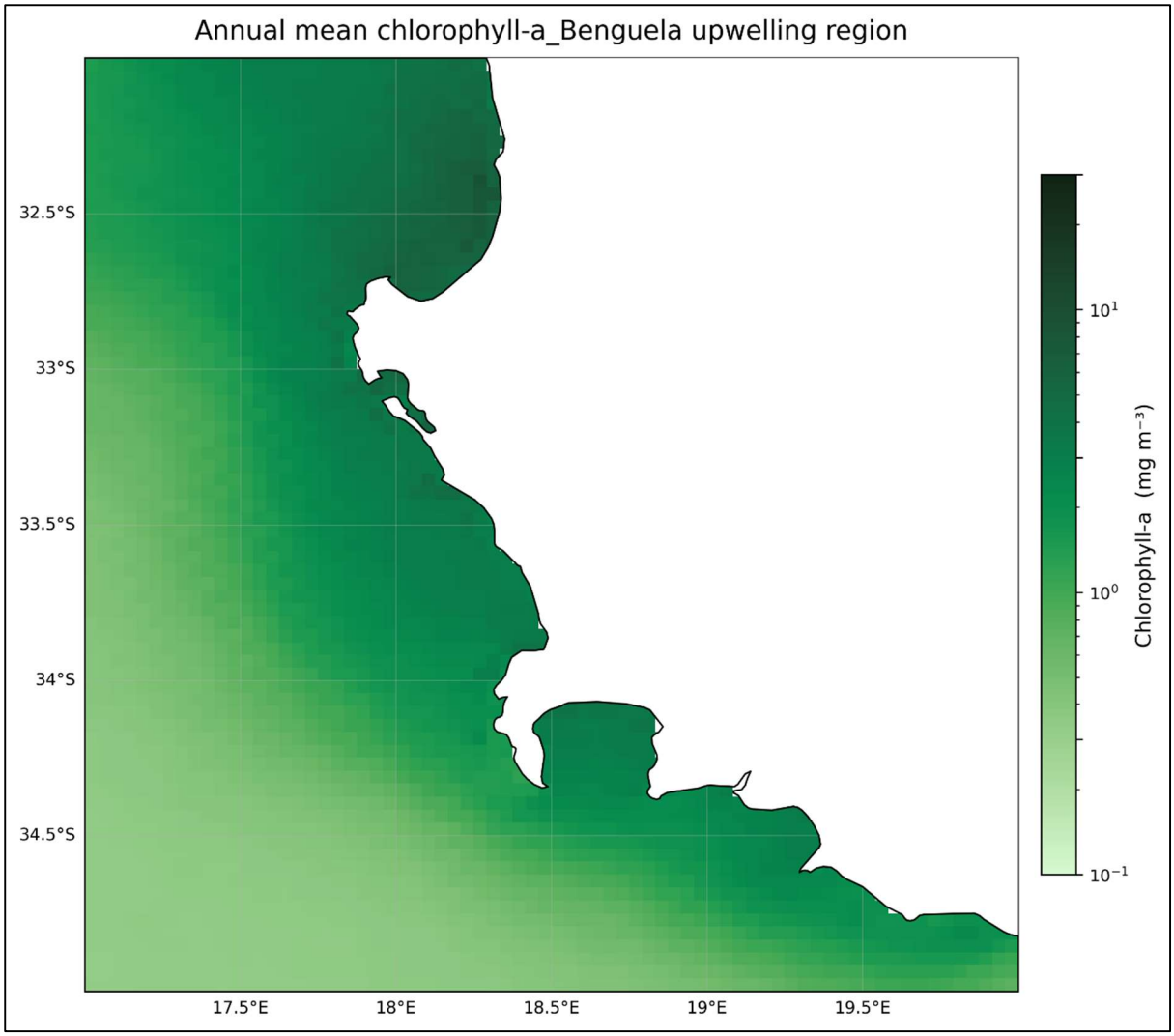

Figure 2: Annual mean chlorophyll-a concentration within the Benguela upwelling system in the southwest coast of South Africa.

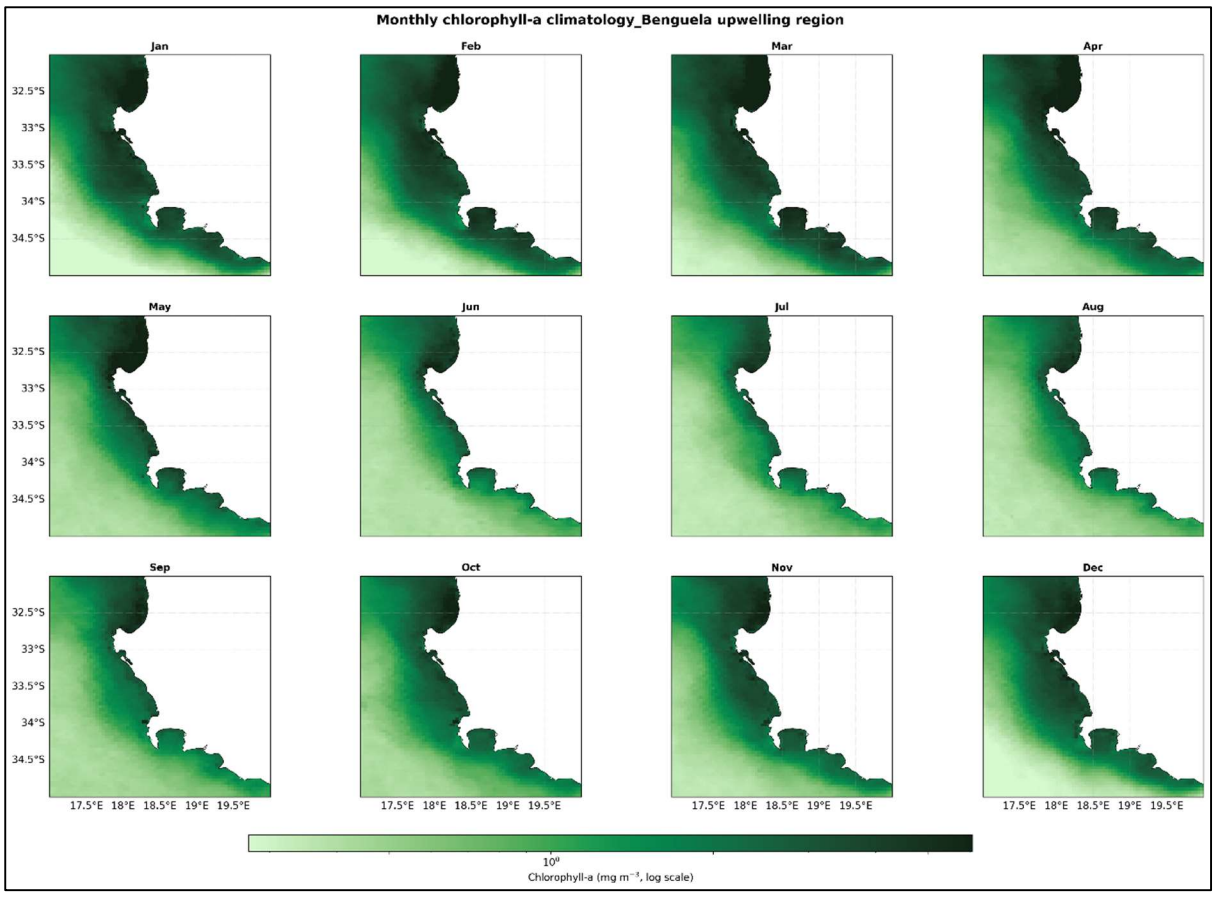

Figure 3: Monthly chlorophyll-a climatology for the study area within the Benguela upwelling system in the southwest coast of South Africa.

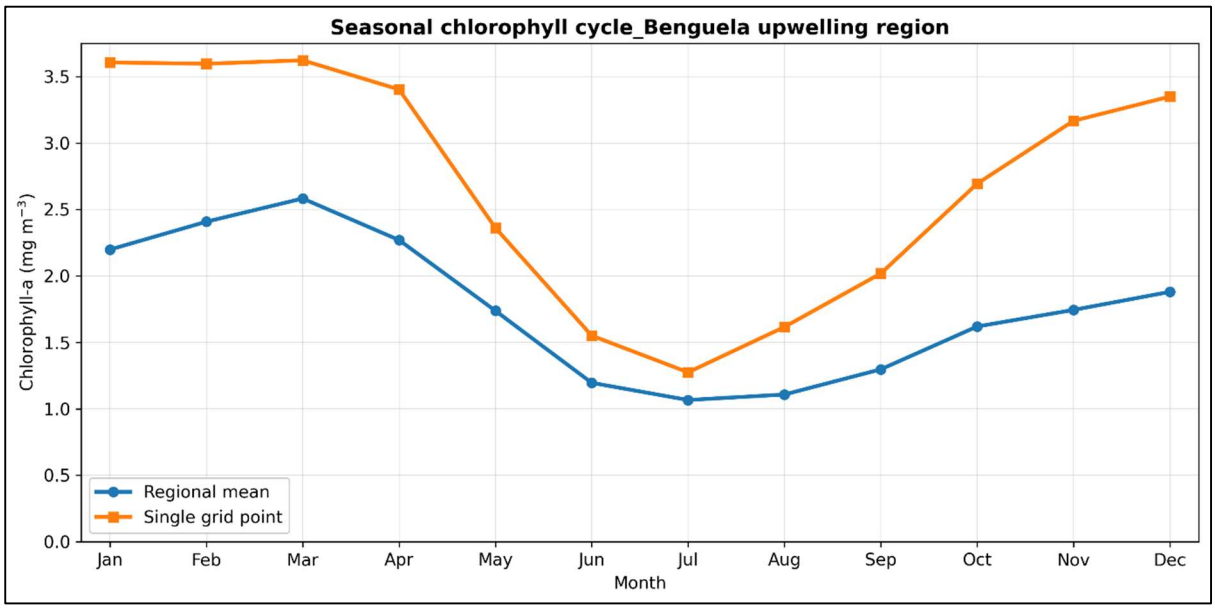

Figure 4: Seasonal chlorophyll-a cycle for the study area and one nearby grid point (-33.8, 18.2).